

MICROCONTROLLER-BASED RING TONE PLAYER

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Mobile phone ring tones sound like real audio recordings. It's not because of the way the melodies are composed, but the protocol behind playing the melody. The ring tone text transfer language (RTTTL) is behind those wonderful lullabies and songs you have on your mobile phone.

Basically, a ring tone is the sound made by a mobile phone to indicate an incoming call or text message. Here we present a microcontroller-based ring tone generator.

The basics

The lowest resonant frequency of a vibrating object is called its fundamental frequency. Most vibrating objects have more than one resonant frequency and those used in musical instruments typically vibrate at harmonics of the fundamental. A harmonic is defined as an integer multiple of the fundamental

frequency.

A cylindrical air column with both ends open vibrates with a fundamental frequency. Each end of the column must be an antinode, with one node at

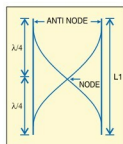


Fig. 1: Fundamental frequency vibration in an open air column

the centre for the air motion. Therefore if ' λ ' is the wavelength of the sound produced by an open cylindrical air column, its length ' L_1 ' will be:

$$L_1 = \lambda/2 (\lambda/4 + \lambda/4)$$

$$\text{So } \lambda = 2L_1$$

Frequency $n_1 = V/\lambda$ (where ' V ' is the velocity of sound)

By putting the value of ' λ ', we get:

$$n_1 = V/2L_1$$

If length is half, then:

$$\text{Frequency } n_2 = V/L_1$$

$$n_1/n_2 = (V/2L_1) \times (L_1/V) = 1/2$$

$$\text{or } n_1 = 2n_2$$

Thus halving the length doubles the frequency, i.e., frequency n_1 is created with one octave higher ($2n_1$).

If the length is made quarter of the original, the frequency becomes $4n_1$, i.e., two octaves higher.

If n_1 is octave 1, then

n_2 is octave 2, then

n_3 is octave 3, and so on.

If the length is doubled, the frequency is halved. That is, the frequency becomes $n_1/2$, which is one octave lower.

In music, frequency n_1 is called a note. The pitch of a piano key or guitar string is described by the note.

According to musical frequency conventions, there are twelve notes in all, namely, A, A#, B, C, C#, D, D#, E, F, F#, G and G#, where '#' sign indicates a sharp note.

According to Nokia RTTTL specifications, note A

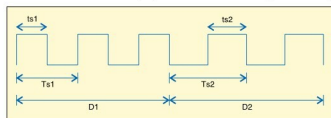


Fig. 2: PWM signal, where Ts_1 , ts_1 and D_1 are the time period, half time period and duration of note 1, while Ts_2 , ts_2 and D_2 are the time period, half time period and duration of note 2

PARTS LIST

Semiconductors:

- IC1 - AT89C51 microcontroller
- IC2 - 7805, 5V regulator
- T1 - BC337 npn transistor
- D1-D4 - 1N4007 rectifier diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 330-ohm
- R2 - 8.2-kilo-ohm
- R3 - 10-kilo-ohm
- R4 - 1-kilo-ohm

Capacitors:

- C1 - 1000 μ F, 25V electrolytic
- C2, C6 - 0.1 μ F ceramic disk
- C3 - 10 μ F, 16V electrolytic
- C4, C5 - 22pF ceramic disk

Miscellaneous:

- X1 - 220V AC primary to 5V, 500mA transformer
- $X_{crystal}$ - 11.0592MHz crystal
- LS1 - 8-ohm, 1W speaker

TABLE I
Nokia RTTTL Frequencies (Hz)
(For Octave 4 to Octave 7)

No.		Octave=4	Octave=5	Octave=6	Octave=7
1	A	220.000	440.000	880.000	1,760.000
2	A#	233.082	466.164	932.328	1,864.655
3	B	246.942	493.883	987.767	1,975.533
4	C	261.626	523.251	1,046.502	2,093.005
5	C#	277.183	554.365	1,108.731	2,217.461
6	D	293.665	587.330	1,174.659	2,349.318
7	D#	311.127	622.254	1,244.508	2,489.016
8	E	329.628	659.255	1,318.510	2,637.020
9	F	349.228	698.456	1,396.913	2,793.826
10	F#	369.994	739.989	1,479.978	2,959.955
11	G	391.995	783.991	1,567.982	3,135.963
12	G#	415.305	830.609	1,661.219	3,322.438